TriboUK 2026

Postgraduate Tribology Conference

1–2 July 2026

University of Sheffield

DAY 1 — WEDNESDAY 1 JULY

MAPPIN HALL

09:00–09:30	Registration
09:30–09:45	Welcome
09:45–10:15	**Keynote 1 — Eladio Hurtado Molina**
10:15–11:15	**Presentation Session 1** *Chair: Alvaro Barrueto* **Yun Zhao** · Imperial College London Lubricating behaviour under electrified conditions **Zhifeng Hu** · University of Huddersfield Condition monitoring of lubricant degradation in journal bearing system based on wireless on-rotor sensing method
11:15–11:30	Coffee break *Sponsored by Phoenix Tribology*
11:30–12:30	**Presentation Session 2** *Chair: Mustafa Faisal* **Sofia Mushtaq** · University of Sheffield Measuring forces experienced on human skin during frictional interactions through a combined approach of digital image correlation (DIC) based strain mapping and machine learning **Zhen Dong** · Imperial College London Laser surface texturing of surgical blades **Osian Thomas** · Ingram Tribology Investigation into the tribological differences between plant and animal proteins using a novel test methodology
12:30–13:10	Travel to AMRC

AMRC

Chair: Forbes Gusha

13:10–13:50	Lunch
13:50–14:00	Introduction to AMRC
14:00–14:30	**Keynote 2 — Leon Proud**
14:30–15:30	**Guided laboratory tours**
15:30–15:55	Networking & closing remarks
15:55–16:40	Travel to University

MAPPIN HALL

16:40–17:50	Drinks reception *Sponsored by PCS Instruments*

GUYSHI

19:00–23:00	Conference dinner

DAY 2 — THURSDAY 2 JULY

MAPPIN HALL

09:00–09:30	Arrival & refreshments
09:30–10:00	**Keynote 3 — Roger Lewis**
10:00–11:00	**Presentation Session 3** *Chair: Charlotte Currie & Jaden Davies* **Song Yang** · Imperial College London Advanced acoustic testing to lubricated contacts **Oluwatamilore Adenipekun** · University of Sheffield An alternative to leaf layer contamination for railhead cleaning trials **Siyu Wang** · Loughborough University Tribological characterization of oil lubricated EV bearing under an electric field
11:00–11:15	Coffee break *Sponsored by Phoenix Tribology*
11:15–12:35	**Presentation Session 4** *Chair: Spike Thompson* **Ziyuan Ren** · Loughborough University The influence of nonlinear friction on transmission error and torsional stiffness in harmonic reducers **Seona Mauchline** · University of Birmingham Interfacial mesh asymmetry governs lubrication and frictional transition in cartilage biomimetic hydrogels **Charlotte Currie** · University of Sheffield Understanding the wear of ceramic abradable liners in jet engine turbine seals **Paula Sebastian Asenjo** · Brunel University of London Optimising radial air foil bearing performance: balancing clearance and stiffness for enhanced lift-off
12:35–13:00	Poster flash talks
13:00–14:00	Lunch & poster session
14:00–14:30	**Keynote 4 — Klaus-Dieter Meck**
14:30–15:30	**Presentation Session 5** *Chair: Damien Dooley* **Musab Rizwan Kazi** · University of Lancashire Additive friction stir deposition (AFSD) for light weight alloys **Uresha Madhuwanthi Herath Mudiyanselage** · Sheffield Hallam University Slurry erosion and erosion-corrosion response of niobium as an alternative cladding material for neutron-producing targets at ISIS **Faraz Shaikh** · University of Leeds Investigation of friction, wear and particle emissions from coated and uncoated commercial vehicle brake rotors using a small-scale tribological approach
15:30–15:35	Voting
15:35–16:00	Coffee break *Sponsored by Phoenix Tribology*
16:00–16:30	Awards & close

Keynote Speakers

KEYNOTE 1

Eladio Hurtado Molina

Lead Engineer, Blade Bearings — Vestas

Eladio Hurtado Molina is a leading expert in wind turbine component reliability, currently serving as Lead Engineer for Blade Bearings at Vestas. After earning his PhD from the University of Sheffield in 2022, he successfully transitioned from small-scale wear research to validating components for the world's largest multi-MW wind turbines. Today, he continues to bridge the gap between industry and academia by managing a strategic collaboration between Vestas and the University of Sheffield.

LP

KEYNOTE 2

Leon Proud

Advanced Manufacturing Research Centre (AMRC)

KEYNOTE 3

Roger Lewis

Professor of Mechanical Engineering, University of Sheffield

Professor Roger Lewis is a Royal Academy of Engineering Research Chair in the Department of Mechanical Engineering at the University of Sheffield, where he has been an academic since 2002. His research spans industrial wear problems, the development of novel ultrasonic techniques for machine element contact analysis, and the design of engineering components and machines. His work covers wheel/rail contact tribology, friction management, condition monitoring, and the tribology of human interactions including skin friction, hand/object contact, and pedestrian slips. He has been recognised by the Tribology Trust Bronze Medal (2001), a Royal Society Brian Mercer Award for Innovation (2003), the Institute of Physics Innovation in Tribology Prize (2008), and the Donald Julius Groen Prize in Tribology (2020).

KEYNOTE 4

Klaus-Dieter Meck

Innovation Director, Core Technology & Simulation — John Crane

Klaus-D. Meck is the Innovation Director — Core Technology & Simulation at John Crane, a division of Smiths Group plc, a global manufacturer and service provider for mechanical seals and supporting equipment. Mr Meck has over 30 years of experience in sealing technology and rotating equipment, with roles in R&D and Application Engineering. He received his master's degree from the University of Stuttgart, Germany and is the author of several papers related to rotating equipment and critical components. He is named inventor on several patents related to mechanical seals, power transmission couplings, and condition monitoring. Mr Meck has recently been appointed as Visiting Professor to the School of Mechanical, Aerospace & Civil Engineering at the University of Sheffield.

Voting

Scan the QR codes or tap the link below to choose your favourites. Voting closes during the final coffee break on Day 2.

Best Oral Presentation

Vote

Best Poster

Vote

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Oral Presentations

Abstracts for the oral presentations, listed in running order.

Lubricating behaviour under electrified conditions

Yun Zhao · *Imperial College London*

As mechanical systems become increasingly electrified, many lubricated components are now exposed to externally applied voltage and current. These electrical stimuli introduce new complexities to lubrication, as interfacial behaviour becomes highly sensitive to electrical conditions. Understanding how lubricants respond under such conditions is therefore essential for ensuring reliable operation and for guiding the design of next-generation lubricant formulations. In this oral presentation, I will summarise our work on lubricant behaviour under electrified conditions. First, I present quartz crystal microbalance (QCM) results for a broad set of additives and rubbing surfaces, both with and without applied voltage, providing a view of additive adsorption phenomena at solid-liquid interfaces. Based on these measurements, I classify lubricant responses to voltage into three categories: cation-anion exchange for ionic additives, spatial molecular orientation for polar additives, and electrochemically driven reactions for a wide range of additive chemistries. The underlying mechanisms and representative results for each category will be discussed. The presentation concludes by outlining the key experimental, theoretical, and engineering challenges that currently limit progress, along with strategies that may accelerate the development of electrified tribology.

Condition monitoring of lubricant degradation in journal bearing system based on wireless on-rotor sensing method

Zhifeng Hu · *University of Huddersfield*

Friction-induced vibrations in journal bearings are typically caused by inadequate lubrication, and lubricant properties play a vital role in preventing such abnormal vibrations. Among numerous lubricant performance indicators, viscosity is one of the most critical factors affecting the stable operation of journal bearings. Therefore, investigating the effect of lubricant viscosity on the friction behavior of journal bearings is fundamentally important. This paper employed a wireless acceleration sensing method to investigate the effects of three lubricants with different viscosities on friction-induced vibrations in a journal bearing system. It was found that higher-viscosity lubricants effectively mitigate high-frequency resonance in journal bearings. Experimental results demonstrate that high-viscosity lubricants can significantly reduce undesirable vibration behavior caused by friction in journal bearings. Additionally, the results validate that the wireless acceleration sensing technique can effectively distinguish the vibration response of journal bearings influenced by lubricant viscosity.

Measuring forces experienced on human skin during frictional interactions through a combined approach of digital image correlation (DIC) based strain mapping and machine learning

Sofia Mushtaq · *University of Sheffield*

Laser surface texturing of surgical blades

Zhen Dong · *Imperial College London*

In surgical blade design, improving blade efficiency and reducing cutting resistance are important challenges, as excessive cutting force and friction during tissue penetration can increase tissue damage and reduce precision. The frictional behaviour of a blade is strongly influenced by its surface properties, particularly at the blade–tissue interface where contact conditions govern cutting resistance. Surface engineering techniques, especially laser surface texturing, have shown potential for modifying frictional behaviour and improving lubrication at contact interfaces.

This project investigates the effect of different laser-generated surface textures on the cutting performance of surgical blades under lubricated cutting conditions. Textured blades were compared with untextured blades through controlled cutting experiments, using PDMS as a tissue analogue and physiological saline as a lubricant. Normal and tangential force measurements were used to evaluate cutting resistance and blade–substrate interaction. Optical imaging with dyed saline was also used to observe lubricant movement around the blade tip and within the cutting region.

The results confirm that laser surface textures can alter blade–material frictional interaction and influence lubricant transport during cutting. Cutting friction is likely influenced by the effective contact area during penetration, as well as by the spacing and position of textures relative to the blade edge. These findings demonstrate the potential of laser surface texturing for improving surgical blade cutting efficiency and provide guidance for future blade surface optimisation.

Investigation into the tribological differences between plant and animal proteins using a novel test methodology

Osian Thomas · *Ingram Tribology*

The demand for the use of alternative proteins has risen dramatically in recent years. They are known to be a more sustainable and resource-efficient solution in comparison to animal-based protein products. However, the sensory profile is unpalatable to the average consumer and requires research to improve mouthfeel. Current tribological test methods are limited both in their abilities to differentiate between small variations, and mimic real-life eating conditions. Factors such as load variation, directionality, lubrication behaviour and biomimicry are vital to aid the development of new test methods on a novel test rig (BTM). Important considerations in this field include more accurately simulating the eating process to enhance the mouthfeel of consumer products, with the aim to improve marketability of more palatable plant-based proteins.

Advanced acoustic testing to lubricated contacts

Song Yang · *Imperial College London*

Acoustic emission (AE) signals provide a non-invasive route for monitoring frictional interfaces, as they contain rich information associated with asperity interactions, lubricant film evolution, and lubrication regime transitions. This study presents an integrated data-driven framework that combines high-frequency AE sensing, signal processing, and machine learning to characterize and predict tribological behavior in lubricated reciprocating sliding contacts. AE signals recorded from high-frequency reciprocating rig tests were segmented into one-second intervals and transformed into frequency-domain features using short-time Fourier transform (STFT). These features were then used to train optimized artificial neural network models for predicting key tribological parameters, including the coefficient of friction, electrical contact resistance, and Stribeck number. By combining the predicted friction coefficient and Stribeck number, complete Stribeck curves were reconstructed from AE data alone, enabling the identification of lubrication regime transitions without direct access to the contact interface. In addition, AE spectral features were used to distinguish lubricant formulations containing different additive chemistries and to classify tribological stages associated with contact evolution and film formation. The proposed models achieved strong predictive performance for frictional and electrical responses, while classification accuracies exceeded 97% across lubricant–stage categories. These results demonstrate that AE signals, when coupled with machine learning, can reveal both quantitative tribological responses and qualitative interfacial states. The framework offers a promising basis for real-time, non-destructive lubrication monitoring, lubricant identification, and predictive maintenance in dynamic mechanical systems.

An alternative to leaf layer contamination for railhead cleaning trials

Oluwatamilore Adenipekun · *University of Sheffield*

Leaf layer contamination, formed by train wheels crushing fallen leaves that mix with oxides and bond to the railhead, causes dangerously low friction at the wheel–rail interface. However, both field-extracted and laboratory-created leaf layers exhibit inconsistent physical properties, such as thickness and adhesion, rendering them unreliable for optimising railhead cleaning technologies. This study evaluates the suitability of paint as a standardised, reproducible proxy for cleaning trials.

Micro-scratch testing was utilised to characterise and compare the scratch hardness, scratch adhesion, and bonding energy of both leaf layer contamination and the paint proxy. Furthermore, field trials were conducted to assess the paint's capability to simulate low-adhesion conditions under operational constraints, using a portable tribometer to measure the coefficient of friction (COF) following a train pass.

Mechanical characterisation revealed that the paint possessed higher scratch hardness and bonding energy than the laboratory-created leaf layer. This establishes a conservative testing baseline, ensuring that cleaning technologies capable of removing the paint proxy will effectively remove actual leaf contamination. Furthermore, field results demonstrated that the paint successfully reduced friction below wet-rail baselines, with a distinct correlation observed between paint drying time and COF. Ultimately, these findings validate this paint formulation as a viable proxy to accelerate the optimisation of railhead cleaning technologies.

Tribological characterization of oil lubricated EV bearing under an electric field

Siyu Wang · *Loughborough University*

Bearings in electric vehicle (EV) powertrains are exposed to coupled mechanical and electrical stresses, which can modify lubricant behaviour and accelerate electrically induced surface damage. This study investigates the tribological performance of a ball bearing under electrified operating conditions using a representative EV lubricant. The results show that EV oil lubrication provides stable mechanical lubrication but does not fully suppress electrical discharge events at the rolling contacts. Sustained discharge activity leads to characteristic surface damage, including localized pitting and progressive degradation of the contact interface. These findings indicate that conventional EV lubricants must satisfy not only friction and wear requirements, but also electrical robustness requirements under realistic powertrain conditions. The study highlights the need for improved EV lubricant formulations with enhanced dielectric control, discharge resistance, and surface protection capability for high-stress electrified bearing applications.

The influence of nonlinear friction on transmission error and torsional stiffness in harmonic reducers

Ziyuan Ren · *Loughborough University*

Harmonic reducers are widely used in robotic and precision transmission systems because of their high reduction ratio, high accuracy and compact structure. However, due to the combined effects of the periodic deformation of the flexspline, tooth meshing, flexible bearing contact and variations in lubrication conditions, the internal friction of the system exhibits significant nonlinear characteristics. This nonlinear friction not only affects transmission error, but also further influences torsional stiffness and dynamic response. To investigate the nonlinear friction behaviour in harmonic reducers, a friction model incorporating Coulomb friction, viscous friction and the Stribeck effect is established and introduced into the analytical framework for transmission error and torsional stiffness. Input and output angular displacement, rotational speed and output torque signals are measured experimentally to extract the transmission error and torsional deformation characteristics under different operating conditions. Frequency domain analysis is then used to investigate the influence of friction nonlinearity on error amplitude, frequency components and stiffness variation. The results show that nonlinear friction has a more significant influence on transmission error at low speed and during speed variation, particularly by causing local enhancement of dynamic error and changes in frequency components. Meanwhile, variations in friction torque and contact state affect the load transmission path, causing the equivalent torsional stiffness of the strain wave gearbox to exhibit nonlinear characteristics. This study contributes to a deeper understanding of the coupling relationship among friction, transmission error and stiffness in strain wave gearboxes, and provides a reference for the modelling, error prediction and performance optimisation of high precision transmission systems.

Interfacial mesh asymmetry governs lubrication and frictional transition in cartilage biomimetic hydrogels

Seona Mauchline · *University of Birmingham*

Osteoarthritis commonly manifests as an isolated cartilage lesion, driven by interfacial degradation during sliding, leading to structural and mechanical asymmetry between opposing seemingly ‘healthy’ cartilage surface. As one surface deteriorates, biphasic lubrication is disrupted, increasing interaction and accelerating degradation within the joint space (Ateshian, 2009). Hydrogels provide a simplified biomimetic model to investigate these mechanisms, mimicking collagen network heterogeneity and localised degradation through mesh size variation (Lin and Klein, 2022). This enables assessment of how mesh mismatch governs lubrication and frictional behaviour, consistent with soft matter lubrication theories in which load bearing is controlled by fluid pressurisation within deformable polymer networks rather than solid–solid contact. A biomimetic hydrogel on hydrogel model was developed to replicate the cartilage interface. Polyacrylamide hydrogels with varied mesh size ratios (ξsubstrate /, ξprobe) ranging from approximately 0.3 to 8.0 (Urueña et al., 2015), were subjected to reciprocating sliding at 0.2 mm/s and 1.0 mm/s. This enabled evaluation of interfacial mesh asymmetry on lubrication and friction under dynamic loading. A tribological transition was observed, in which the coefficient of friction (μ) increased as mesh size ratio increased. Closely matched systems ((ξsubstrate / ξprobe) < 1.5) exhibited a low-friction regime (μ typically < 0.05), consistent with sustained fluid load support. In contrast, increasing mesh asymmetry raised friction in a mesh-size dependent manner, with transitions occurring at higher ratios (ξsubstrate/ ξprobe > 4). At high asymmetry, friction increased (μ > 0.05), indicating a shift from fluid-supported lubrication to polymer network interactions and viscoelastic dissipation. Higher sliding velocity further amplified friction, particularly at elevated mesh ratios, suggesting velocity dependent destabilisation of the hydrated interfacial region responsible for load-bearing lubrication. These findings highlight that minimising interfacial mesh mismatch is key to maintaining low friction and provide a framework for cartilage repair design, where structural compatibility governs long-term durability.

Understanding the wear of ceramic abradable liners in jet engine turbine seals

Charlotte Currie · *University of Sheffield*

Abradable seals, used in jet engine compressors and turbines, minimise clearance between blade tips and the engine casings, reducing gas leakage around blades. This increases engine efficiency, reducing fuel consumption and emissions.

To achieve this, abradable seals employ sacrificial liners inside engine cases. During running and handling, when the engine is first turned on, the blades rub this lining, which wears in preference to the blades, creating a tight clearance groove to run in during standard operation. Adverse conditions, such as thermal cycling or turbulence, result in further rubs. Ideally, the liner continues wearing in preference to the blade, ensuring clearances remain low after such events. Predictability and control over such rubs is limited, requiring seals to perform across a wide range of conditions.

This research experimentally investigates the behaviour of existing turbine sealing systems under a range of conditions, where ceramic abradables are used to withstand high temperatures, in combination with c-BN blade tips. Existing research shows these tips fail under extreme conditions, leading to blade wear and damaging the seal’s performance. Whilst tip breakdown is well understood, the abradable wear that leads to it requires further investigation in this study. Two primary findings are observed. Firstly, the abradable's dominant wear mechanism changes with increasing incursion rate, limiting the blade's ability to keep up with the increased material removal required, consequently leading to system failure. Secondly, the system breaks down as incursion depth and rub duration increase. Material transfers from the abradable to the blade tip during the rub, reducing the grit's protrusion and cutting capability. This increases the contact forces and temperatures, resulting in further material transfer and grit wear, and eventual system failure. Understanding this will inform future systems that encourage abradable breakdown and extend the blade's ability to cut despite material transfer, ultimately preventing system failure.

Optimising radial air foil bearing performance: balancing clearance and stiffness for enhanced lift-off

Paula Sebastian Asenjo · *Brunel University of London*

Air Foil Radial Bearings (AFRB) are mainly used to support shaft weight, vibrations loads and to prevent the compressor and turbine wheel from rubbing against the housing. From a tribological point of view, the friction created by the air gap and the wear of the foils are investigated in order to reduce the tip clearances of the wheel. The objective is to create the thinnest air film gap possible with the best acceptable losses and overall, still gain aerodynamic efficiency. Different clearances are obtained by changing the foil geometries. The ultimate goal is to test the bearings for lift-off and high speeds of up to 100k rpm, achieving minimal power losses and maximising net aerodynamics efficiency of the overall machine. AFRB are widely applied in high-speed turbomachinery that can run up to 100k rpm like micro gas turbines and air compressor. The challenges of reducing the compressor wheel's tip clearance are that the bearing suffers more losses, higher friction and at high speeds it can overheat leading to ultimate failure. Thus, the results of this study will give guidance on how to reduce the bearing clearance but still perform in accordance to the customer's compressor. A purpose-built centrifugal test rig, capable of reaching speeds up to 24k rpm will be presented. This is followed by a discussion on how the apparatus operates to measure friction for different rotational speeds and data is logged. In the obtained start-stop cycles of speed against friction, the friction measured before, during, and after the bearing's lift-off will serve as key performance indicators. The final results will be presented to show how different foil geometries affect stiffness and lift-off performance of the bearing. Finally, the bearings undergo high-speed testing of up to 100k rpm to evaluate the aerodynamic performance of the bearings with different clearances. This project in as example on how tribology can be studied in air foil bearing to improve the overall efficiency of real-life machines.

Additive friction stir deposition (AFSD) for light weight alloys

Musab Rizwan Kazi · *University of Lancashire*

A new solid-state additive manufacturing method called Additive Friction Stir Deposition (AFSD) makes it possible to deposit metallic materials below their melting points, which lowers flaws like porosity and hot cracking. The aim of this study is to understand the process and optimise the deposition quality using numerical and experimental methods. The study will focus on processing lightweight alloys (Al, Mg, and Ti) using AFSD. Even though AFSD has many benefits for applications in the automotive, marine, and aerospace industries, there are still difficulties associated with the process due to the various parameters that affect the quality.

Slurry erosion and erosion-corrosion response of niobium as an alternative cladding material for neutron-producing targets at ISIS

Uresha Madhuwanthi Herath Mudiyanselage · *Sheffield Hallam University*

Slurry erosion and erosion-corrosion response of Niobium as an alternative cladding material for neutron-producing targets at ISIS H. M. L. U. Madhuwanthi1, Y. P. Purandare1, A.P. Ehiasarian1, A. Dey2, L. Jones2, S. Gallimore2 1 School of Engineering and Built Environment, National HIPIMS technology centre, Sheffield Hallam University, City Campus, Howard Street, Sheffield, S1 1WB, United Kingdom 2 Rutherford Appleton Laboratory, Didcot, OX11 0QX, United Kingdom

Abstract

The TS2 spallation targets at ISIS comprise a Tungsten (W) core clad with a thin Tantalum (Ta) sleeve. Gamma-spectral analysis of the target cooling-water loop has detected radionuclides including 172Lu, 175Hf, 187W, and 182Ta, suggesting a materials-related constraint in the current target configuration that contributes to premature failure. Among the proposed degradation mechanisms, erosion–corrosion of the Ta cladding is considered a significant contributor, in addition to thermal loading and radiation-induced damage. Owing to its comparable chemical and physical properties, Niobium (Nb) is being considered as a candidate cladding material. The present work focuses on the erosion–corrosion behaviour of Nb (99.8 wt.%) using a bespoke impinging jet erosion–corrosion apparatus. Aqueous slurry erosion experiments were performed at impingement angles of 20, 30, 45, 60, and 90°, at a constant slurry concentration of 7 wt.% SiC particles in de-ionised water and a jet velocity of 6 ms-1. The erosion rate exhibited a maximum value of 4.52×10^(-6) kgm-2s-1 at an impingement angle of 30°, decreasing to 1.21×10^(-6) kgm-2s-1 at 90°, consistent with erosion behaviour characteristic of ductile metals. To assess the combined effect of erosion and corrosion, experiments were extended to corrosive environments under both active dissolution and passivating potentials, and in the presence or absence of abrasive particles. Three corrosive media were examined: pH 3 and pH 5 (0.5 M citric buffer solutions) and pH 7 (3.5 wt.% NaCl solution). As anticipated, the total material loss rates in erosion–corrosion processes (Kec) were consistently higher than those for pure erosion (Keo) and pure corrosion (Kco) alone. The trend of higher material loss at shallow impingement angles, particularly at 30°, persisted regardless of pH or potential. Furthermore, the average material loss ratios Kec / (Keo + Kco) were 0.2, 0.8, and 1.2 at pH 7, 5, and 3, respectively, indicating an increasing synergistic erosion-corrosion effect with increasing solution acidity. At pH 3, erosion–corrosion was dominated by an accelerated removal of the passive film due to particle impact combined with enhanced chemical dissolution of Niobium whereas at pH 5 and 7, protective passive layers mitigated particle-induced damage.

Investigation of friction, wear and particle emissions from coated and uncoated commercial vehicle brake rotors using a small-scale tribological approach

Faraz Shaikh · *University of Leeds*

Brake systems are a significant non-exhaust source of airborne particulate emissions, particularly in urban environments where regenerative braking is not always sufficient to eliminate friction braking events. While a large proportion of existing research has focused on passenger vehicles, there remains limited understanding of how commercial vehicle brake materials behave under demanding operating conditions involving higher loads, temperatures, and prolonged braking events.

This study investigates the tribological behaviour and particle emissions of coated and uncoated commercial vehicle brake rotors using a small-scale experimental approach. A laser-cladded titanium carbide (TiC) coated rotor was compared with a conventional grey cast iron (GCI) rotor under controlled laboratory conditions using a Bruker UMT TriboLab tribometer. Testing was based on a modified WLTP-style braking procedure adapted for small-scale operation. Friction behaviour, temperature response, wear characteristics, and airborne particle emissions were evaluated throughout the testing programme.

Particle measurements were carried out using a Dekati ELPI+ system, while post-test surface analysis was performed using optical microscopy and surface characterisation techniques to assess wear mechanisms and transfer layer development. Initial findings show clear differences in friction stability, surface condition, and emission behaviour between the coated and uncoated rotor materials.

The work demonstrates how small-scale tribological testing can be used to investigate environmentally relevant brake wear behaviour while reducing the cost and complexity associated with full-scale dynamometer testing. The study also contributes towards the wider understanding of how advanced rotor coatings may influence future low-emission braking systems for commercial vehicles.

Poster Presentations

Abstracts for the poster presentations.

How to replenish lubricant in an oscillating wind turbine blade bearing raceway

Alvaro Barrueto · *University of Sheffield*

Wind turbine blade bearings connect the blades to the hub and allow them to pitch (rotate) according to varying wind speed. Blade pitching generates small-amplitude oscillations that push grease out of the contact and generate lubricant starvation. This research project examines raceway surface wear generated under blade bearing conditions, and which preventive strategies can be developed to mitigate raceway damage.

Microstructure-controlled mixed lubrication by WPI-stabilised water-in-water emulsions

Chenxi Wang · *University of Leeds*

This study investigates the lubrication behaviour of dextran/polyethylene oxide water-in-water (W/W) emulsions stabilised by whey protein isolate (WPI) (D–W–P) as a model platform for aqueous lubricants in biomedical applications. Using soft-contact tribology under physiologically relevant loads and entrainment speeds, we compared WPI-stabilised emulsions with their protein-free W/W counterparts and with non-phase-separated solutions. Systems containing WPI showed substantially lower friction coefficients in the mixed-lubrication regime than their protein-free counterparts, demonstrating that WPI provides an additional friction-reducing contribution. Among all formulations, the lowest friction was obtained when WPI was combined with a phase-separated W/W microstructure. Increasing dextran concentration, corresponding to a higher (dextran-rich) droplet volume fraction, led to a further reduction in friction, consistent with more pronounced droplet accumulation and restructuring in the inlet region. In contrast, low-viscosity WPI solutions with matched WPI content provided only modest lubrication benefit, indicating that WPI alone is insufficient in the absence of viscous support from the bulk phase. Overall, the results suggest a synergistic mixed-lubrication mechanism, in which the viscous P-rich continuous phase provides load support, while WPI stabilises D-rich droplets under shear and thereby promotes more persistent inlet accumulation, increasing the local resistance to flow and enhancing entrainment. These findings offer a microstructure-guided route for designing aqueous lubricants for soft tribological interfaces.

The future of dry gas seal materials

Damien Dooley · *University of Sheffield*

Dry gas seals are dry running, non-contacting seals that limit leakage from a pressurised system with a rotating component by the formation of a small, stable fluid film gap between opposing face seals. These seal faces are not in contact during typical running, but during start up and slow down, will come into contact and experience a friction event.

Future dry gas seal technology will require a higher density of start-stop cycles and as such, new wear resistant face seal materials capable of surviving more friction events are necessary. My research aims to produce novel silicon carbide matrix composites containing novel transition metal-based solid lubricants by spark plasma sintering (SPS).

My current research is investigating the successful synthesis of the matrix-lubricant pairings and compositions for future sample production. This is paired with physical and tribological characterisation by performing SEM, XRD, friction testing and wear analyses.

Ultrasonic measurement of lubricant film thickness in ball-raceway contacts of rolling element bearings

Jaden Davies · *University of Sheffield*

Real-time wear-corrosion measurement using eddy current and electrochemistry sensing

MM Raihan · *University of Leeds*

Biomedical CoCrMo alloys, although considered passive, can undergo corrosion in the human body under tribological contact (e.g., hip and knee joints), a process known as tribocorrosion [1]. This results in greater material loss than wear or corrosion alone. Current preclinical methods (ISO 14242-2) primarily assess wear using mechanical simulators, requiring test interruption and extensive post-processing, which limits real-time evaluation of degradation mechanisms. In-situ assessment techniques provide continuous monitoring without interrupting testing, enabling better understanding of the interaction between mechanical and electrochemical processes. This study aims to evaluate the tribocorrosive degradation of CoCrMo alloys using in-situ techniques, specifically eddy current sensing and electrochemical measurements. The approach is expected to improve the accuracy of degradation prediction and support the development of more reliable biomedical implants.

Minimised delubrication of legume proteins by enzymatic hydrolysis

Mingxin Wang · *University of Leeds*

With the expanding scale of the plant-based market, plant proteins have been widely incorporated into various formulations. However, undesirable sensory attributes, particularly astringency, of plant proteins remain the major barrier to broader consumer acceptance. Astringency has been found strongly associated with high friction as measured in tribological experiments. Hence, we demonstrated a biochemical approach, i.e. enzymatic hydrolysis to reduce oral friction of plant protein dispersions by using different tribology models. Oral friction of samples was measured using a Rheo-tribometer equipped with 3D printed biomimetic tongue-like surfaces. We found hydrolysis effectively reduced boundary frictions, which have been considered relevant to astringency by up to two-fold, while remained viscosity constant. Analysis of protein particle size and surface wettability suggests that this reduction was mainly associated with decreased particle size and improved surface wetting ability. To further investigate the roles of bulk saliva and the salivary pellicle in plant protein lubrication, bovine submaxillary mucin (Mucin) was mixed with legume proteins or its hydrolysates, and the mixtures were tested on an in-vitro model salivary pellicle. A pronounced delubrication was induced by intact legume protein addition on mucin pellicle. In contrast, hydrolysed samples caused substantially less delubrication, exhibiting friction values close to those of the lubricious mucin pellicle. This distinct behaviour is attributed to displacement of mucin by intact legume protein through competitive surface adsorption, whereas hydrolysates enabled co-adsorption with mucin, as supported by theoretical modelling (Self-consistent field). These findings advanced the understanding of plant protein delubrication and offered a feasible approach to formulating next-generation plant-based products with improved mouthfeel.

The effect of lubricant contamination on fretting wear of engine piston rings and cylinder liners in PHEVs

Mu Qiao · *University of Leeds*

Plug-in hybrid electric vehicle (PHEV) operation is characterised by frequent stop–start events and short engine-on periods, creating a uniquely challenging lubrication environment in which bulk oil temperature may remain suppressed and the evaporation of volatile contaminants is limited. Consequently, fuel dilution and water can accumulate and persist in the lubricant. An on-road fleet study conducted under extreme cold conditions reported fuel dilution up to 20% (m/m) and water dilution approaching 5% in PHEV engine oils [1]. During engine-off intervals, many interfaces receive no lubricant replenishment and are sustained only by residual oil films that may already contain elevated fuel and water concentrations. Vehicle-borne vibration can then impose small-amplitude micro-motion on nominally static contacts under boundary-dominated and potentially starved conditions, thereby creating potential conditions for fretting-type degradation [2]. Despite these risks, the combined effects of fuel dilution and water-in-oil emulsification under fretting conditions remain insufficiently understood. This study therefore quantifies how PHEV-relevant fuel dilution and water-in-oil emulsification affect friction, wear, and zinc dialkyldithiophosphate (ZDDP) tribofilm formation under lubricated fretting conditions, with reciprocating sliding used as a benchmark. This investigation builds directly on prior evidence that water can degrade ZDDP anti-wear performance [3], fuel can alter the tribological response of ZDDP-containing lubricants beyond viscosity effects [4], and simultaneous gasoline–water dilution can suppress film thickness and antiwear film development while increasing wear risk [5]. Experiments are conducted using a ball-on-flat ferrous contact using the tribometer shown in Figure 1. The lubricant formulation comprises PAO-6 blended with secondary ZDDP at 800 ppm phosphorus (P), with contamination matrices of water (0.5–3 wt%) and gasoline (5–20 wt%). Fretting tests are performed at 30 N, 30 Hz, and ±10 µm for 3 h. Sliding benchmarks on a TE77 are conducted at 30 N, 5 Hz, and ±5 mm for 3 h at 80 °C.

Development of PFAS-free ionic liquid greases for tribological applications

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Conventional lubricating greases often contain per- and polyfluoroalkyl substances (PFAS), which incorporate halogen-containing compounds known for their excellent thermal stability and ability to reduce friction and wear in mechanical systems. However, the persistence, resistance to degradation, and potential environmental impacts of PFAS have raised significant concerns regarding their long-term sustainability. As a result, halogen-free ionic liquids have emerged as promising alternatives for grease formulation. Despite their potential, studies investigating halogen-free ionic liquids as grease additives remain limited, and their application as base oils for grease formulation has received even less attention. In the present study, a halogen-free ionic liquid was synthesized and evaluated as a potential base oil for the development of a new generation of environmentally sustainable greases. The synthesized ionic liquid was blended with selected concentrations of a thickener to formulate a novel grease. The chemical structure and composition of the ionic liquid were characterized, and the influence of drying temperature on its structure stability was systematically investigated. The results provide valuable insights into the reaction pathway, chemical composition, and drying temperature stability of the synthesized halogen-free ionic liquid. These findings will contribute to a better understanding of ionic liquid-based grease formulations and support the development of sustainable lubrication technologies for future tribological applications.

Evaluating the friction recovery and damage behaviour of sand alternatives in the wheel-rail contact

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Each autumn, a great number of leaves fall off trees onto rail tracks. The wheels of a train can crush the leaves which fall on the railway, forming a 'black layer' on the surface. This layer reduces friction levels between rails and wheels, which can cause delays and even accidents. To solve this problem, sanding is currently one of the main methods used to improve adhesion, but the resources of high-quality railway sand are limited, and there is an urgent need to find alternative materials. This paper studies four potential alternative materials: steel grains, chilled iron, alumina and garnet, and evaluates their effects on restoring adhesion and their impact on wheel-rail wear. High-Pressure Torsion machine (HPT) was used to simulate the wheel-rail contact area with leaf layer contamination and the coefficient of friction was measured in the test zone. Scanning Electron Microscopy (SEM) was used to analyse the shape of the four particles and the surface of HPT samples after the test.

Fabrication of photoluminescent thin films via tribochemical deposition of upconversion nanoparticles

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Micro- and nano-scale manufacturing enables the creation of high-precision structures for diverse applications in optics, electronics, biomedical devices, and sensors. Tribochemistry, which harnesses friction-induced interfacial reactions, provides a promising route for film fabrication at these scales. Among various functional materials, upconversion nanoparticles (UCNPs) have gained significant attention over the years due to their unique characteristics, including anti-Stokes shifts, sharp emission peaks, resistance to photobleaching, and long luminescent lifetimes. In this study, lanthanide-doped UCNPs were investigated as tribofilm precursors for the fabrication of functionalized tribofilms responsive to near-infrared (NIR) light at both the micro- and nano-scale. Micro-scale fabrication experiments demonstrated that UCNPs could be successfully incorporated into iron oxide-based tribofilms after extended sliding, while retaining their luminescent properties. Secondary Ion Mass Spectrometry (SIMS) analysis revealed a layered tribofilm structure, with UCNPs predominantly concentrated in the upper and middle regions. At the nanoscale, in-situ AFM experiments showed that stable UCNPs tribofilm formation required the presence of iron oxide, which also account for the film's limited thickness and high susceptibility to load-induced wear. This study demonstrates a viable strategy for integrating optical functionality into tribofilms, opening opportunities for tribochemistry-driven fabrication in sensing and micro/nanomanufacturing applications.

Probabilistic distribution of ball-raceway contact force and contact pressure in the blade pitch bearing of the IWT7.5 reference wind turbine

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Wind turbine (WT) blade pitch bearings operate under loads that vary with wind speed, leading to variations in the localised ball-raceway contact forces and contact pressures. Commonly used methods for contact analysis often rely on deterministic approaches, which do not account for the probabilistic variations in contact force and contact pressure. To overcome the limitations, this study proposes a probabilistic framework for evaluating the probability distributions of maximum contact force and maximum contact pressure in the blade pitch bearing of the IWT7.5-164 reference WT, with weighting based on wind speed occurrence. Publicly available IWT7.5-164 external bearing loads are represented by axial force F_a, radial force F_r, and resultant moment M_res. Weibull wind speed probability distributions are established using wind speed data from the Anholt offshore wind farm. A Hertzian-contact-based analytical bearing model, solved using the Newton-Raphson method, is implemented to determine the nonlinear equilibrium between external bearing loads and localised ball-raceway contact forces. Wind speed weighted probabilistic distributions of contact force and contact pressure are determined. The results show that large contact pressure events occur with the highest probability within the wind speed interval of 10 m/s to 12 m/s, which includes the rated wind speed of 11.7 m/s. The maximum contact pressure reaches 2679.36 MPa, occurring at a wind speed of 11.28 m/s under Design Load Case (DLC) 1.1. Under the nominated wind speed Weibull distribution at a height of 120 meters, contact pressures in the range of 2546.47 to 2679.36 MPa account for 0.30% of the analysed DLC 1.1. Similarly, contact pressures in the range of 2261.02 to 2313.63 MPa occur most frequently, contributing 8.94%. Compared with deterministic contact load and contact pressure analysis, the proposed framework identifies not only the magnitudes of contact forces and contact pressures, but also their probabilities of occurrence with limited computational cost.